

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

(0250402)

I Bhuvaneswararaj (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naive Bayes / Logistic Regression / SVM), or Reinforcement Learning applications.*

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

- (a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.
- (b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

- (a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.
- (b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

- (a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.
- (b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?
- (c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

- (a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.
 - (b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.
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RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

Deep Reinforcement Learning Approach for Trading Automation in The Stock Market.

Article Summary:

Authors: Taylan Kabbani and Ekrem Duman

Journal: IEEE Access

Volume: 10

Year: 2022

Domain: Artificial Intelligence | Automated Stock Trading

ML Problem: The problem is to develop an autonomous trading agent that can decide how to allocate capital among stocks and perform buy/sell actions while considering market conditions, transaction costs, technical indicators and news sentiment.

Objective:

The main objective of the article is to develop an automated stock-trading system using Deep Reinforcement Learning (DRL). The authors formulate stock trading as a Partially Observed Markov Decision Process (POMDP) and use the TD3 algorithm to learn an optimal trading policy.

Methodology Analysis:

Machine Learning Technique:-

The primary technique used in the article is:

TD3 Algorithm;

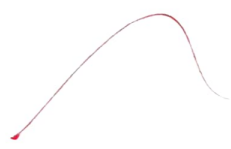
It is a Deep Reinforcement Learning algorithm and is an improved version of Deep Deterministic Policy Gradient.

The author selected TD3 because stock trading has a continuous action space. Instead of restricting the agent to only:

- Buy
- Sell
- Hold

Dataset used:

The financial dataset contains:

- (i) Date
 - (ii) Volume
 - (iii) Open Price
 - (iv) Close Price
 - (v) Adjusted Close
 - (vi) High Price
 - (vii) Low Price
- 

Methodological Strength:

The major strength is the use of continuous action space. This makes the model more realistic for portfolio allocation because the agent is not restricted to simply buying, selling, or holding a fixing number of shares.

Results and Critical Evaluation:

The experiments were averaged across five independent runs / random seeds. The baseline Sharpe ratio was 1.43, compared with 0.85 for the benchmark used in the study.

Sharpe ratio = 2.68

Total profit = \$110,368

Critical Evaluation:

The results are promising, but they should be interpreted carefully. A Sharpe ratio of 2.68 on unseen data is strong, and the authors describe it as very good / good depending on the section of the paper.

The authors attempted to improve reliability by:
using multiple random seeds

Testing on unseen market

Comparing different state representation.

Possible biases:

Historical - data bias

Market period bias

Asset selection bias

Hyperparameter sensitivity

News sentiment limitations

Real world Applicability:

Automated Trading:

The agent could automatically determine portfolio allocation.

Portfolio Management:

Financial institutions could be used as Reinforcement Learning System to assist with asset allocation.

Financial Decision Support:

The system could provide recommendations to traders based on market information.

Algorithmic Trading:

The approach could be incorporated into algorithmic trading system that continuously analyze market condition.

Reflection and Learning Outcome:

Insight 1:

I learned that reinforcement learning differs from conventional supervised learning because the agent learns by interacting.

Insight 2:

The article helped me understand why continuous action spaces can be useful in real world application as decision making.

Insight 3:

The result showed that adding useful information to the state can improve the agent's performance.

Proposed Novel Experiment:

I would extend this research by testing the TD3-based trading system on the Indian stock market, using the stock such as those from the NIFTY 50 index.

Expected impact:

This experiment would determine whether the proposed reinforcement learning approach generalizes beyond the market used in the original study.

Overall Critical Review:

The article is highly relevant to demonstrates reinforcement learning in a real world decision making problem. The authors formulate automated trading as a POMDP and use TD3 to learn an optimal trading policy. The experimental results are encouraging. Adding technical indicators increased the sharpe ratio to 2.75, while adding sentiment information increased it to 3.14 in the corresponding evaluation. However, the result should ^{not} be interpreted as a guarantee of future performance. Financial markets change over time, and historical back testing. Overall, the article provides a clear practical answer / example how RL can be used for automated decision making.